

		Initial $\frac{1}{2}mv^2$
3	A	At steady speed of 1.4 m s^{-1}, air resistance = weight component along slope $1.4 k = (80.0)(9.81)\sin 5^\circ$ $k = 48.857$ At steady speed of 5.5 m s^{-1}, air resistance = $(5.5)(48.857) = 268.71 \text{ N}$ Additional force required = $268.71 - (80.0)(9.81)\sin 5^\circ = 200 \text{ N}$ (2 s.f.)
4	B	$x = 10.0, 7.5, 2.5 \text{ cm}$